

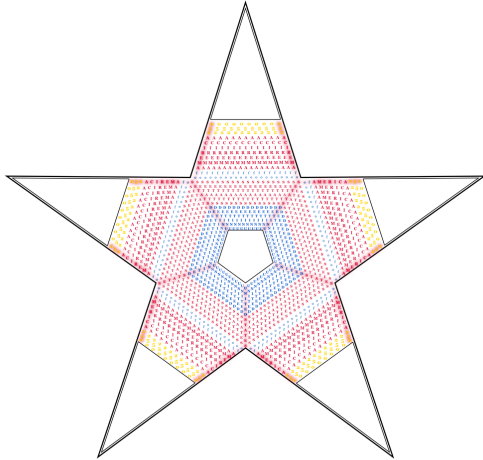
THE GREAT AMERICAN STAR

BIRTHDAY PROBLEM - EDUCATOR GUIDE

The Great American Star "Birthday Problem"

Educator Resources · Probability Exploration

How quickly does the chance of a matching path grow as more students participate? This lesson borrows the idea behind the classic Birthday Problem and applies it to the **359,277,250 possible paths** in The Great American Star.



About the mathematical setting

The Great American Star is a mathematical-art project created for America 250. For this probability lesson, each of its **359,277,250 paths** is treated as one possible choice.

The question: If students independently choose paths at random, how likely is it that at least two students choose the same path?

Core comparison

The number of student-pairs can be exactly the same while the chance of a match changes dramatically because the pool of possible choices is radically different.

1

pair with 2 students

190

pairs with 20 students

365

possible birthdays

359,277,250

possible star paths

The key idea

Two quantities matter: **(1) how many pairs can be formed**, and **(2) how large the pool of possible choices is**. The lesson is designed to make students discover both.

1. Learning goals and teaching sequence

Learning goals

- Count how many distinct pairs can be formed from a group of students.
- Distinguish between the number of comparisons and the size of the possibility pool.
- Compare the classic Birthday Problem with path matching in The Great American Star.
- Understand why an exact "at least one match" probability is most naturally found by calculating the probability of no matches first.
- Make predictions, explain reasoning, and interpret extremely small probabilities in context.

Recommended classroom setup	Use one teacher computer connected to a projector or smartboard, or printed copies of the activity. Students can do the probability reasoning on paper. Individual devices are not required. If devices are available, the Star Creator can be used after the lesson so students can choose a path of their own.
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Our mathematical model

For the Great American Star calculation, assume every path is equally likely and each student chooses independently. This model describes the mathematics; it does not claim that real students using the creator will choose paths uniformly. For the classic birthday comparison, assume 365 equally likely birthdays, independent from student to student, and ignore leap day.

Suggested teaching sequence

- 1. Start with two students:** Ask students to estimate how likely an exact path match should be when there are 359,277,250 possible choices.
- 2. Increase to twenty:** Before calculating probability, ask how many student-pairs there are. This reveals why counting pairs is essential.
- 3. Find 190 pairs:** Build or derive $20 \times 19 \div 2 = 190$, emphasizing that Alex-Sam and Sam-Alex are the same pair.
- 4. Compare with birthdays:** Keep the same 20 students and the same 190 pairs, but change the pool from 359,277,250 paths to 365 birthdays.
- 5. Reveal the exact probabilities:** Use the no-match method, then subtract from 1.
- 6. Discuss the deeper idea:** Ask students which quantity is driving the difference: number of pairs, size of the choice pool, or both.

Vocabulary

Pair	Two students being compared.
Possibility pool	The total number of choices that could be selected.
Independent choices	One student's choice does not change another student's probabilities.
At least one match	One or more pairs in the group choose the same outcome.

2. From one pair to 190 pairs

Start with two students

Suppose two students each independently choose one of the 359,277,250 Great American Star paths at random. The first student may choose any path. For the second student to match the first, exactly **one** of the 359,277,250 paths works.

Chance of the same path	1 in 359,277,250	Approximately 0.000000278%
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What changes with 20 students?

Every student can potentially match every other student. The key is to count **pairs**, not simply multiply the two-student probability by 20.

$20 \times 19 \div 2$ $=$ **190 pairs**

Each of the 20 students can be paired with 19 others. That counts every pair twice: Alex-Sam and Sam-Alex are the same pair. Dividing by 2 counts each pair once.

20 students produce 190 different student-pairs. That means there are 190 opportunities for a match.

At least one matching pair	0.0000529%	Roughly 1 chance in 1.89 million
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Teaching point	The number of pairs has grown dramatically - from 1 to 190 - but 190 opportunities are still spread across a pool of 359,277,250 possible paths. A duplicate remains extraordinarily unlikely.
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3. Exact probability: calculate no match first

For teachers: the exact 20-student calculation

It is easier to calculate the probability that *no one matches*, then subtract that result from 1.

$$P(\text{no match}) = (359,277,250 / 359,277,250) \times (359,277,249 / 359,277,250) \times \dots \times (359,277,231 / 359,277,250)$$

$$P(\text{at least one match}) = 1 - P(\text{no match})$$

Exact result

The product gives approximately **0.0000528839366%** for at least one duplicate anywhere in the 20-student group. On the student-facing material, this is rounded to **0.0000529%**.

Why not just add 190 pair probabilities?

Multiplying the single-pair probability by 190 gives an extremely close approximation here because matches are so rare. But it is not the exact probability of "at least one match": a group can contain more than one matching pair, so the pair events can overlap. The no-match product handles those overlaps correctly.

General pair-count formula

For n students:

$$n(n - 1) / 2 \text{ pairs}$$

This formula is a useful bridge to combinations: the number of unordered pairs is "n choose 2." The lesson can remain accessible without introducing combination notation, or the notation can be added for older students.

4. Same students. A much smaller pool.

The classic birthday comparison

Now keep the same students and the same pair counts, but replace 359,277,250 possible paths with 365 possible birthdays. Assume 365 equally likely birthdays, independent from student to student, and ignore leap day.

Group	Student-pairs	Possible choices	Birthdays: match chance	Great American Star: match chance
2 students	1	365 vs. 359,277,250	0.274%	0.000000278%
20 students	190	365 vs. 359,277,250	41.1%	0.0000529%

SAME 190 PAIRS.

VERY DIFFERENT SIZE OF THE POSSIBILITY POOL.

20 students create 190 possible pairs whether we are comparing birthdays or Great American Star paths. What changes dramatically is the size of the pool: **365 birthdays** versus **359,277,250 paths**.

- **More students** -> many more pairs -> more chances to match.
- **A much larger pool of possibilities** -> a much smaller chance that any pair actually matches.

The surprising comparison	For 20 students, the classic birthday problem reaches about 41.1% , while the Great American Star model remains at only about 0.0000529% . The pair count is identical; the possibility pool is not.
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5. Questions, short answers, and extensions

Questions to ask students

Q: Why does going from 2 students to 20 students create 190 pairs instead of only 20?

A: Because every student can be paired with every other student. Counting each unordered pair once gives $20 \times 19 \div 2 = 190$.

Q: Why does the probability grow so dramatically in the birthday problem?

A: The number of pairs rises to 190 while the pool remains only 365 birthdays, so many comparisons are competing inside a relatively small set of possibilities.

Q: Why does the Great American Star probability remain tiny?

A: The same 190 pairs are spread across 359,277,250 possible paths.

Q: Which matters more here: the number of students or the number of possible choices?

A: Both matter. More students create more pairs, while a larger choice pool makes each exact match less likely.

Optional extension: 30, 50, and 100 students

These values use the same independent, equally likely model. They can be revealed after students make predictions.

Students	Pairs	Birthday: at least one match	Star path: at least one match
30	435	70.6%	0.000121%
50	1,225	97.0%	0.000341%
100	4,950	99.99997%	0.001378%

The key idea

The deeper lesson is that **two quantities matter**: how many pairs can be formed, and how large the pool of possible choices is. The classic birthday problem has many pairs competing inside a small pool of 365 possibilities. The Great American Star has the same pairs spread across 359,277,250 possible paths.

Connect back to the student experience

After the probability lesson, students can open the Star Creator and choose a path of their own. The mathematics has just shown how enormous the possibility space is; the creator turns that abstract number into a personal choice.

359,277,250 possible paths. One American star. This one is mine.